PROJECT ECO-BASKET PITCH DECK

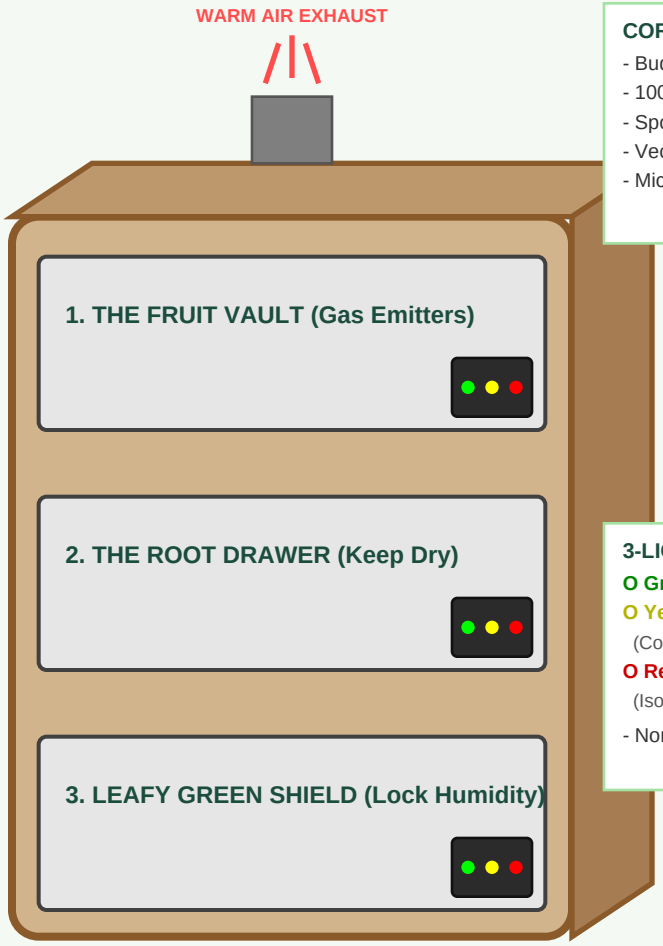
A Two-Page Formal Engineering and Economic Solution Blueprint

PROJECT ECO-BASKET

THE GRID-INDEPENDENT FRESHNESS BOX FOR URBAN COMMUNITIES

HOW IT IS MADE

- Pressed Rice Husk Chassis (Insulated)
- Food-Grade Stainless Steel Drawers



CORE ADVANTAGES

- Budget Optimizer
- 100% Grid Independent
- Spoilage Chains Cut
- Vector Shield Design
- Microclimate Isolation

HOW IT WORKS

- Zero Grid Power
- Warm Air Convection
- Automated Bi-Metallic Gas Vent Flaps
- Low Air Intake Draw

3-LIGHT STATUS SYSTEM

O Green: Optimal Fresh
O Yellow: Impending Rot
(Consume Immediately)
O Red: Spoilage Alert
(Isolate Compartment)
- Non-Electric Sensor

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TECHNICAL EXPLANATION & VIABILITY REPORT

1. System Architecture & How it is Made

Project Eco-Basket changes standard household food storage by building a completely passive, sealed microclimate containment system. The physical envelope is constructed as an enclosed box structure to maximize environmental insulation and shield vulnerable fresh items from open-air contamination vectors.

- **Outer Chassis Composition:** The structurally isolated outer shell is manufactured entirely from pressed agricultural rice husks (agro-waste). By heating and highly pressing the raw fibrous husk matrix with bio-resins, a rigid structural panel with high thermal insulation capacity is obtained. This diverts waste streams into value-added hardware.
- **Internal Containment Drawers:** Slide-out compartments are manufactured from grade-304 medical food safe stainless steel. This ensures zero chemical leaching, highly non-porous clean surfaces, high corrosion resistance to ambient wet environments, and rapid thermal conductivity to distribute passive temperature gradients cleanly.

2. Thermodynamics & Operational Principles

The system functions entirely without external energy inputs or grid connection, utilizing foundational fluid physics principles to preserve fresh produce continuously.

- **Convective Ventilation Loop:** Biological respiration releases warm metabolic vapors. The box structure creates a natural chimney effect, directing the warm, buoyant, gas-rich air upward to vent autonomously through a low-resistance top throat.
- **Automated Gas & Temperature Regulation:** Custom-tuned bi-metallic mechanical vent flaps are situated along the upper thermal escape channels. As internal heat indexes rise or moisture-carrying gas concentrations spike, the differential expansions of the mechanical metal bindings force the flaps open automatically without an electronic controller.

3. The Three-Light Freshness Status Module & Day Tracking

To bring highly sophisticated tracking information to user profiles without grid reliance, each storage compartment incorporates an advanced, standalone low-draw sensing assembly powered by a compact, high-efficiency mechanical kinetic micro-cell.

- ■ **Green LED Pulse (Optimal Atmosphere):** Indicates that gas concentration values, volatile compounds, and moisture profiles are well within the safe biological storage parameters.
- ■ **Yellow LED Pulse (Impending Rot Window):** Flags initial detection of rapid organic breakdown or high concentrations of ripening gases. This provides an explicit daily life prompt to the household to consume the target produce item immediately, effectively preventing rot from taking over the stack.
- ■ **Red LED Pulse (Spoilage Alert):** Alerts users to advanced cellular failure or high mold spore concentration. This prompts the user to completely pull out the sliding compartment and isolate the spoiled

items, preventing cross-contamination chains.

• **Manual Day-In-Storage Indexer:** Integrated linear mechanical dials (1-7 days) provide low-cost, human-centric inventory management tailored perfectly for users in high-density communities.

4. Economic Viability & Daily Profitability

Metric Evaluated	Standard Storage	Eco-Basket System	Net Financial Benefit
Monthly Budget Food Spoilage Waste	30% - 35% Decay Rate	< 8% Target Loss	Saves over 2,500 INR / Month
Grid Power Operational Overhead	Variable Volt Cost	0.00 INR (Passive)	Insulates against blackouts
Chassis Asset Life-cycle Cost	Short Life Plastics	Highly Durable Shell	Low depreciation amortized

By lowering produce spoilage by over 25%, the unit provides measurable economic leverage to low-income populations, acting as a secure financial shield for daily household survival.

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